
QT 408: Introduction to Quantum Materials

Course Code	QT 408
Course Name	Introduction to Quantum Materials
L-T-P-C	3-0-0-3
Prerequisites	Engineering Mathematics (Linear Algebra, Complex algebra, basics of 2nd of ODEs and initial value problems, 2nd order PDEs and boundary value problems, Probability and Statistics, Random variables). Maxwell's equations and EM theory at the level of the core physics syllabus from AICTE model
Intended for	UG/PG/PhD
Distribution	Elective PG/Elective UG
Semester	Odd/Even
Mutual Exclusion	PH 523, EP 301
Approval	47th Senate (Sep 2025)

Preamble

Students of this course learn (i) The basic idea of quantum materials, (ii) The basics of band theory of solids, (iii) The basics of magnetism, (iv) The basics of superconductivity, (v) About new 2D materials like graphene, TMDCs, (vi) About topology and topological phases of matter

Course Contents

- **Band theory basics**
 - Metals, Semiconductors and Insulators
 - Band structure of solids
 - Survey of semiconducting devices for quantum technologies (electronic, quantum optical devices and principle of operation)
- **Correlated systems**
- **Magnetism**
 - Para, ferro magnetism basics
 - Magnetic measurements, hall effect, magnetoresistance
 - Faraday and Kerr effects
- **Superconductivity**
 - BCS theory
 - Ginzburg Landau
 - Josephson Effect – AC and DC Josephson effects
 - Survey of superconducting devices for quantum technologies
- **2D materials**
 - Graphene and its properties – single and few layers
 - Transition Metal Dichalcogenides – Electronic and Optical Properties
- **Topological Phases of matter**
 - Basics of Topology
 - Geometric phases - Berry Phase

- Aharonov Bohm effect
- Topological phases of matter
- **Survey of material growth techniques**
 - Molecular beam epitaxy
 - Chemical vapor deposition, MOVPE
 - Pulsed laser deposition, etc.
 - Crystal growth techniques

Course Outcomes

Students of this course learn

1. The basic idea of quantum materials
2. The basics of band theory of solids
3. The basics of magnetism
4. The basics of superconductivity
5. About new 2D materials like graphene, TMDCs
6. About topology and topological phases of matter

References

1. *Condensed Matter Physics*, M P Marder, 2nd Edition, John Wiley and Sons, 2010
2. *Introduction to Superconductivity*, Michael Tinkham, standard ed., Medtech (2017)